

FROZEN YET ADAPTIVE: SINGLE-EPISODE REPAIR OF VISION–LANGUAGE–ACTION POLICIES FROM PROPRIOCEPTIVE RESIDUALS

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ABSTRACT

A vision–language–action (VLA) policy is a large frozen network whose training assumed a healthy robot. When the execution hardware drifts—a miscalibrated tool frame, a stiff or weakened joint—the policy is silently wrong, and fine-tuning it is out of reach for most deployments. We show that the fault can be identified and cancelled *within a single episode*, from the robot’s own motion, by adapting six numbers at the policy’s action interface: a plant model and a sensitivity matrix identified once on healthy data turn the proprioceptive residual into an estimate of the fault, and a projected, normalised gradient law drives it to zero. No gradients through the policy, no reward, no reference trajectory; on tasks whose margin is smaller than the estimator’s motion, the estimate identified in one to three episodes is held thereafter. On four LIBERO suites with a frozen $\pi_{0.5}$, paired episodes go from 23% to 65% success (exact McNemar $p = 2.4 \times 10^{-14}$, $n = 120$), and from 28% to 57% when the calibration is identified on initial states the evaluation never visits, with every suite individually significant either way; across 2,240 paired episodes in 110 stored cells (faults, severities, profiles, controls and ablations, tallied by a script from the stored outcomes) the correction fixes 625 and breaks 66. The same law repairs two further backbones from two other developers (OpenVLA-OFT, NVIDIA GR00T N1.7) with the *same* calibration, and, in joint space with a calibration of their own from a few healthy rollouts, two further manipulators: a bimanual ALOHA and a humanoid (Fourier GR1 under GR00T N1.5: on two tasks, 3% $\rightarrow$ 63% and 3% $\rightarrow$ 43%, no regression in sixty paired episodes). Matched estimators that share the plant model repair as well as this law; a baseline without it reaches 35% where the method reaches 95%: the contribution is the calibrated interface, not the update rule. We state the two conditions the evidence supports—identifiability, and a reparability criterion comparing the task’s spatial margin against the correction’s variation—and the failure modes that bound the method, including faults invisible to proprioception and a joint lock that is identified but cannot be cancelled from the action interface.

1 INTRODUCTION

Vision–language–action models map images and a language instruction to robot actions and are trained on demonstrations collected from a particular robot in a particular state of repair. Deployed, they are frozen. The robot is not: tool frames drift, calibrations go stale, joints lose effectiveness. Each changes the map from the policy’s action to the robot’s motion, and the policy, which never compares commanded motion with achieved motion, has no mechanism to notice.

The standard remedies are to fine-tune on faulted data, which requires labelled faults and the compute to update billions of parameters, or to learn a residual correction offline against reference rollouts. Both need the fault in a dataset before it can be corrected: the closest contemporary method, J-PARC (Jo et al., 2026), trains its calibrator per environment from faulted rollouts against fault-free references. Neither is available to a robot that degrades while running.

This paper takes the classical control view. A fault at the action interface is a disturbance on a plant whose healthy dynamics can be identified once; the disturbance is observable in the difference

between the motion the healthy plant would have produced and the motion actually measured; and an adaptive law with a handful of parameters can cancel it online. The question is whether this survives a 3.35B-parameter policy, contact-rich tasks, and a residual that carries model error as well as the fault. Our contributions:

1. **A single-episode repair loop for frozen VLAs** (§4) needing only a plant model and a sensitivity matrix identified on healthy data, both properties of the robot rather than of the policy or the task. On LIBERO the estimate resets every episode and reconverges within tens of steps; where the controller keeps its target it is identified in one to three episodes and held.
2. **Evidence at scale, paired** (§6): four suites, three backbones, four robots in two simulators, four fault families, three severities and four time profiles, evaluated episode-for-episode against the frozen policy with exact McNemar tests, a healthy control on every cell, and 2,240 paired episodes in aggregate, tallied by script.
3. **Two conditions, stated as propositions** (§5): a constant input fault and a constant output bias are not separable within one episode, so the healthy bias must be measured outside it; and repair succeeds when the correction’s within-episode variation fits inside the task’s spatial margin, both quantities being measurable before any repair attempt. The second predicts every outcome on three further robots, including the ones continuous adaptation fails.
4. **The boundary** (§7): what the method cannot see (sensor and camera faults), cannot fix (a policy that fails unfaulted), and gets wrong (one healthy episode in twenty on one backbone, a sustained phantom).

2 RELATED WORK

Where adaptation enters the stack. Generalist policies map observations to actions by token prediction (Zitkovich et al., 2023; Kim et al., 2025b), diffusion (Chi et al., 2023; Ghosh et al., 2024) or flow matching (Black et al., 2025b;a); OFT studies efficient fine-tuning (Kim et al., 2025a). All improve the policy; our problem is a deployment change in the map from its actions to motion.

Adaptive augmentation of a pretrained policy. Cheng et al. (2022) wrap a reinforcement-learning policy in $\mathcal{L}_1$ adaptive control, with hardware; DATT adds an $\mathcal{L}_1$ disturbance estimator to a learned trajectory controller (Huang et al., 2023). Residual policy learning trains an additive action (Silver et al., 2018; Johannink et al., 2019); RMA trains the policy to consume an adapted context (Kumar et al., 2021). Our adapter estimates a disturbance from calibrated proprioceptive residuals and leaves the VLA untouched; the architecture is classical, the target is not.

Residual calibration of frozen VLAs. The closest work is J-PARC (Jo et al., 2026): joint-level faults on frozen $\pi_{0.5}$ and OpenVLA-OFT across the same LIBERO suites, repaired by a calibrator trained *offline*, per environment, from faulted rollouts with fault-free references and teacher targets. We use no faulted data, no references and no teacher: the plant is identified once on healthy motion and the fault online in the episode it occurs in. The channel mask of our headline cells was first found by a faulted separation test and is reproduced by a gate on the healthy phantom alone (§4.3); hyperparameters remain development-tuned. Their faults enter below the controller, ours at the action interface; §6.5 tests their class with our law. Closed-loop activation editing (Das et al., 2026) steers a frozen policy through its activations; our correction needs no access to them.

Estimation and adaptive control. The law is a disturbance observer (Shim, 2020) with feedforward cancellation in the MRAC gradient form, with the standard robustness modifications (Ioannou & Sun, 1996); analytical redundancy (Isermann, 2005; Chen & Patton, 1999) and composite adaptation (Slotine & Li, 1989) supply the tools. Neural-Fly (O’Connell et al., 2022), HMAc (Xie et al., 2024) and MAGIC-VFM (Lupu et al., 2025) learn disturbance representations and adapt their coefficients online. What is new here is not the estimator but its target: a frozen generative policy whose command distribution decides what is identifiable, and a task whose spatial margin decides what is repairable. Matched observers sharing our plant model repair as well as the law does (§6.4), which is the point.

3 SETTING

A frozen policy π_ω emits an action chunk from which the controller consumes $a_t \in \mathbb{R}^d$ per step. For LIBERO and the WidowX a_t is a six-dimensional Cartesian increment plus gripper, executed by the benchmark’s controller; for ALOHA it is fourteen and for the GR1 humanoid twenty-nine absolute joint targets executed by position servos. The measured state y_t is the end-effector increment or the joint position.

A fault corrupts the executed action:

$$a_t^{\text{exec}} = a_t + f \quad (\text{additive}), \quad a_t^{\text{exec}} = \text{diag}(g) a_t \quad (\text{multiplicative}),$$

constant or varying in time. The fault is injected between the policy and the controller. For LIBERO this models a miscalibrated tool frame or a stale hand-eye calibration; it is *not* a joint-level actuator fault, which would enter below the controller where the fault-to-motion map is state dependent (§8).

The policy never sees f . It sees images and the proprioceptive state, and $\pi_{0.5}$ -LIBERO ignores the state input entirely (zeroing it changes the action by exactly 0.00000), so a perception-side fault cannot be constructed for it nor an action-side fault corrected by it.

4 METHOD

4.1 PLANT, RESIDUAL, AND SENSITIVITY

On healthy rollouts we identify a per-channel FIR plant

$$\hat{y}_t = \sum_{\ell=0}^K h_\ell u_{t-\ell} + c, \quad K = 6, \quad (1)$$

where u_t is the command we *believe* we sent (the policy’s action plus any correction we applied). For LIBERO the plant predicts the increment; for ALOHA it predicts the position, because there the command is an absolute target that a servo tracks and a target offset lives in the steady-state position error, not in the increment (§6.7). The residual

$$r_t = y_t - \hat{y}_t \quad (2)$$

is what the healthy plant cannot explain. Because u_t includes the correction and the world executes $u_t + f$, $r_t \approx Mf$ independently of the correction, so the residual estimates the *total* fault. $M = \partial(\text{motion})/\partial(\text{fault})$ is a $d \times d$ sensitivity matrix measured once by open-loop replay: a recorded healthy command sequence is replayed with and without a per-axis probe, and the mean motion difference per unit probe gives each column. On the LIBERO plant M is diagonal-dominant with $\text{cond}(M) = 3.0$ and is magnitude independent to within 14% between probes of 0.02 and 0.05; on ALOHA $M \approx I$ with $\text{cond}(M) = 7.3$; on the GR1 the arm block is $0.99 I$ by a direct step response from a held pose (the replay probe is contaminated on an arm that touches objects during the replay).

4.2 THE ADAPTIVE LAW

Per control step,

$$e_t = M^{-1}r_t - b, \quad \tilde{e}_t = \frac{e_t \mathbb{I}[\|r_t\| > \delta]}{1 + \|r_t\|^2/\rho^2}, \quad \hat{f}_{t+1} = \Pi_{[-\kappa, \kappa]}(\hat{f}_t + \gamma(\tilde{e}_t - \hat{f}_t)), \quad (3)$$

and the next action is sent as $a_t - \hat{f}_t \odot m$, where m selects the channels to correct. The terms are the standard robustness modifications and each earned its place by fixing an observed failure: the dead-zone δ stops plant noise integrating into drift, the normalisation damps outlier steps (contacts, joint limits), the projection κ keeps the estimate in a physical box, and b is the healthy-plant phantom—the value $\hat{f}$ settles at with no fault present—measured offline and subtracted, exactly as a sensor is zeroed before it is trusted (Proposition 1 says why it cannot be fitted within the episode).

Remark 1 (Constants are ratios of measured scales). *Every threshold in equation 3 is compared against a quantity with a scale: δ against the residual norm of the healthy plant, ρ against the residual*

norm under the fault, κ against the expected fault, the excitation gate of §4.4 against the regressor's magnitude. Seven silent failures in this work came from carrying a constant tuned on one plant to another—a deadzone larger than the residual it gated, a normaliser that attenuated every ALOHA update to 6%, a normaliser fed the residual of joints the law was not estimating, a projection bound that coincided with the true fault and was read as a measurement. The rule that follows is mechanical: print the scale, then set the constant as a multiple of it, on every new plant.

The norms in equation 3 are taken over the channels being corrected: on the humanoid the twelve unmodelled hand joints carry three times the arm's residual, and a norm over all joints attenuated every arm update until the estimate settled at 13% of the fault. The fixed point of equation 3 is attenuated by $1/(1 + \|Mf\|^2/\rho^2)$; at the measured $\|r\|$ on LIBERO this is 5%, matching the observed 2% shortfall, and an innovation-normalised variant ($e_t = M^{-1}(r_t - M\hat{f}_t)$, unbiased fixed point) was indistinguishable there (0.12σ) at twice the variance. We keep equation 3 on LIBERO and ALOHA and use the innovation form on the humanoid, where the fault is three times the healthy residual and the bias would be 50%.

4.3 WHICH CHANNELS TO CORRECT

A gate on the healthy phantom alone (three across-episode standard deviations of the estimate on healthy episodes, re-evaluated every step) reproduces the mask and the headline cell, $9/20 \rightarrow 19/20$, 10 fixed, 0 broken, with a healthy control of $20/20 \rightarrow 19/20$ (Appendix A). The separation test—the estimate under a fault minus the estimate on matched healthy episodes—is the only evidence of identification we accept; a raw $\hat{f}$ carries the plant phantom and cannot distinguish identification from bias. For the uniform six-axis fault used across the LIBERO suites, rotation identifies (+0.049 against a true 0.050) and translation does not (sign-wrong), so the headline correction is restricted to rotation. That restriction is a property of *that fault*, not of the channel: on a translation-only fault the same law identifies translation at 79%–106% and repairs it (§6.3). The uniform fault differs because single-axis responses stop superposing at ± 0.05 , the signature of a contact or joint limit reached only when all six axes move.

4.4 MULTIPLICATIVE FAULTS

For a gain fault the regressor follows from the plant rather than being assumed. The world executes gu and the FIR responds to what it is given, so per channel

$$y_i - \hat{y}_i = \beta_i \phi_i, \quad \phi_i = \hat{y}_i - c_i, \quad \beta_i = g_i - 1, \quad (4)$$

the regressor being the FIR-weighted command history, and the regression lives in motion units with no M^{-1} . Estimating β_i by recursive least squares on channels whose $|\phi_i|$ clears an excitation gate, the correction is the exact certainty-equivalent inverse $a_i/(1 + \hat{\beta}_i)$ with $\hat{g}_i$ projected away from zero. Two facts matter here. The compensator *inverts* the estimate, so estimation error is amplified by $1/\hat{g}^2$ —fourfold at $g = 0.5$ —and accuracy sufficient for an additive fault is not sufficient for a multiplicative one. And a wrong regressor is not repaired by adding parameters: regressing on the instantaneous command instead of ϕ_i produced a 33% phantom loss of effectiveness on a *healthy* robot, an intercept made it worse, and the derived regressor shrank the phantom ninefold and took the x -channel separation from 1.4σ to 18σ .

5 TWO CONDITIONS

Proposition 1 (A constant input fault and a constant output bias are not separable). *Let $y_t = M(a_t + f) + b + v_t$ with M square and full rank, and f, b constant over the episode. For any b' , the fault $f' = f + M^{-1}(b - b')$ reproduces every observation exactly. Only $Mf + b$ is identified.*

Proof. $M(a_t + f') + b' = Ma_t + Mf + (b - b') + b' = M(a_t + f) + b.$ $\square$

Consequently the healthy phantom b must come from outside the episode (we measure it on fault-free rollouts), and a fault that switches on mid-episode is *better* posed than one present from the start: b is identified before onset and f after.

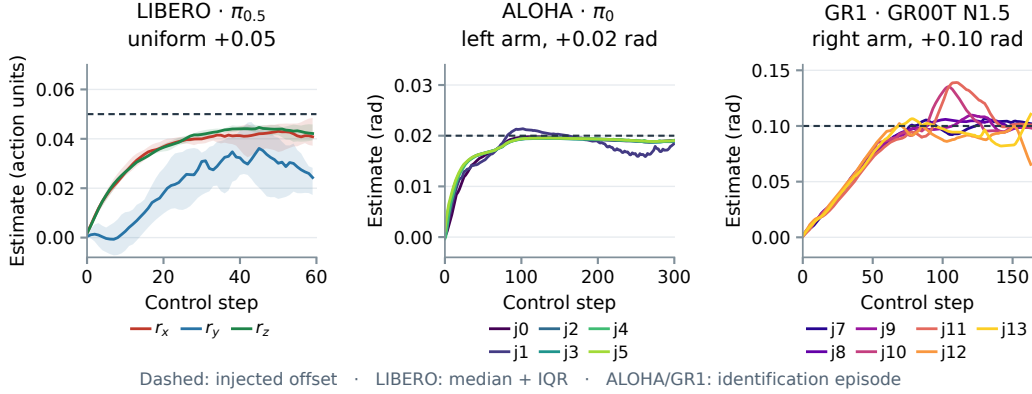


Figure 1: The fault estimate within one episode, from zero, on three robots. Left: LIBERO spatial, $\pi_{0.5}$, uniform +0.05 fault, rotation channels (median and interquartile range over 20 episodes): two reach 0.04 by step 60, the third 0.024, all still rising; at episode end the separation test puts them at 0.047–0.049. Middle: ALOHA transfer-cube, π_0 , +0.02 rad on the left arm’s six joints; the identification episode settles at 92–96% of the offset, then held (§6.7). Right: GR1 humanoid, GR00T N1.5, +0.10 rad on the seven right-arm joints; 90–112% within 100 steps, then held (§6.8).

Proposition 2 (Repairability). *Let μ be the task’s spatial margin—the largest steady residual offset at the end-effector under which the healthy policy still succeeds at its healthy rate—and let σ be the within-episode variation of the applied correction. Repair recovers the healthy rate when the residual stays within μ for the whole episode, for which $\sigma < \mu$ is necessary. Both μ (by a static-correction sweep) and σ (from the estimator’s trajectory on one faulted episode) are measurable without knowing the fault and without a repair trial.*

This is an empirical proposition, not a theorem: it states the quantity the data turned out to depend on. It predicts LIBERO (a reach with centimetres of margin; σ of 0.019 cm per step; repair), ALOHA with continuous adaptation ($\mu < 0.5$ cm; $\sigma = 0.43$ cm; no repair), ALOHA with the held estimate ($\sigma = 0$; repair) (§6.7), and the GR1 humanoid both ways on two tasks (§6.8).

6 EXPERIMENTS

6.1 PROTOCOL

Paired design. Both arms of every cell run the same (task, initial state) episodes and differ only in whether the correction is applied; the policy’s own action sampling is not pinned across arms, so a fixed or broken episode is an observed paired difference, not a counterfactual; Figure 1 shows what the corrected arm does inside one episode. We report exact McNemar tests on the discordant pairs; unpaired tests on these data are conservative and, worse, hide the regression count. **Controls.** Every cell has a healthy-frozen control on the same episodes: a null against a ceiling (frozen already near 100%) or a floor (frozen near 0 unfaulted) is not a null; four cells in this work were avoidable with that control first. **Noise.** $\pi_{0.5}$ samples its action flow; the identical condition scored 13/20, 13/20, 15/20 on three runs, so ± 10 points at $n = 20$ is free variation and we quote intervals. **Calibration.** Plant and M were identified once on `libero_spatial` healthy rollouts with $\pi_{0.5}$ driving and are used unchanged for every LIBERO suite and for OpenVLA-OFT; ALOHA has its own from eight healthy rollouts, the GR1 from six, the WidowX from ten. The shipped M was probed at initial state 45, which the headline cells evaluate; the preregistered recalibration on disjoint states is the right-hand column of Table 1 and Appendix D.

6.2 FOUR SUITES

Uniform +0.05 offset on all six action dimensions, rotation-only correction (§4.3), $\pi_{0.5}$ -LIBERO.

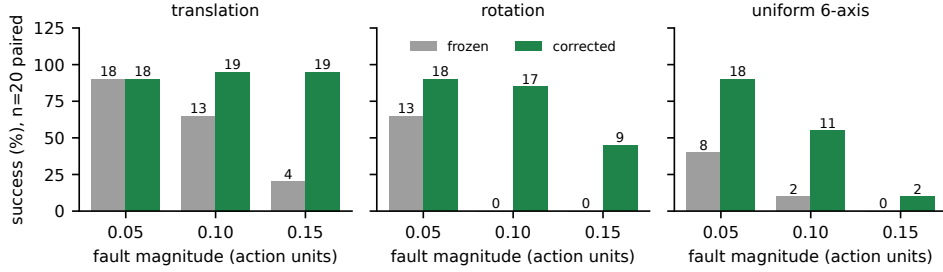


Figure 2: The recoverability map: frozen versus corrected success on LIBERO spatial, $n = 20$ paired episodes per cell, correction on the channels each family’s separation test supports (rotation for the rotation and uniform families, translation for translation). Every regression sits in a ceiling cell (translation 0.05 is 1 fixed, 1 broken behind its equal totals). Single-family faults degrade gracefully with magnitude; the uniform fault collapses at 0.15 because its uncorrected translation component drives the arm into contacts and limits (§7). Exact McNemar p -values are in the table below.

suite	n	calibration probed at an evaluated state					held-out calibration	
		frozen	corrected	fixed	broken	McNemar	frozen	corrected
libero_spatial	20	8/20	18/20	10	0	0.0020	9/20	18/20
libero_goal	40	15/40	29/40	15	1	0.00052	18/40	28/40
libero_object	20	5/20	16/20	11	0	0.00098	7/20	15/20
libero_10	40	0/40	15/40	15	0	6.1×10^{-5}	0/40	7/40
pooled	120	28/120	78/120	51	1	2.4×10^{-14}	34/120	68/120

Every suite is individually significant in both columns (held-out: 0.0039, 0.0063, 0.039, 0.016; 37 fixed, 3 broken pooled). The single regression in the left column is one of the 28 episodes the frozen policy was winning (3.6%, the denominator that can regress). The right column answers an objection the left invites: the shipped M was probed at initial state 45, which these cells evaluate. Re-identifying plant and M at states from 25 upward leaves three suites within one episode and costs the long-horizon suite half its repair, 15/40 $\rightarrow$ 7/40 (preregistered; the drop refutes the registered bound of five, Appendix D). A 20% error in M ’s translation block has two hundred steps to accumulate on libero_10 and forty on the others: the cost of calibrating away from the deployment state. libero_90, on which the checkpoint was never fine-tuned, scores 6/20 unfaulted and is reported as a competence floor of the policy, not a repair result.

6.3 FAULT FAMILIES AND SEVERITY

Figure 2 and the table give the fault-family by magnitude map (two cells at $n = 40$ in Appendix H; constants ablation in Appendix E).

fault (libero_spatial, $n = 20$)	0.05	0.10	0.15
translation	18/20 $\rightarrow$ 18/20 (ceiling)	13/20 $\rightarrow$ 19/20	4/20 $\rightarrow$ 19/20
rotation	13/20 $\rightarrow$ 18/20	0/20 $\rightarrow$ 17/20	0/20 $\rightarrow$ 9/20
uniform 6-axis	8/20 $\rightarrow$ 18/20	2/20 $\rightarrow$ 11/20	0/20 $\rightarrow$ 2/20

One regression, in the ceiling cell. Rotation is where this policy breaks—a rotation fault at 0.10 takes it to exactly zero where translation leaves 13/20—and the correction recovers 17/20 from that floor. Single-family faults degrade gracefully with severity; the uniform fault collapses, and the comparison of the last column isolates why: the rotation fault and the uniform fault are corrected identically, and adding an *uncorrected* translation component drops recovery from 45% to 10%. The boundary is the plant leaving the regime where its motion is informative, not the estimator. A structured mixed-sign fault $[0, 0, 0, +.06, -.06, +.03]$ the law was never tuned on goes 2/12 $\rightarrow$ 10/12 with separation matching truth on every axis.

Time-varying faults (amplitude 0.10): a ramp to full over 60 steps goes $12/20 \rightarrow 18/20$ ($p = 0.031$); a non-zero-mean oscillation on rotation $13/20 \rightarrow 20/20$ ($p = 0.016$); an intermittent fault $12/20 \rightarrow 19/20$ ($p = 0.039$). The estimator tracks the waveform poorly—45–79% RMS error, not attributable to lag—and it does not matter: the correction must be roughly right in direction and scale, not a good tracker, which is why six numbers updated on CPU suffice. (A zero-mean oscillation averages away and does no damage; that cell proved nothing and is not counted.)

Loss of effectiveness (§4.4): at $g = 0.30$, $1/20 \rightarrow 17/20$ ($p = 3.1 \times 10^{-5}$) with $\hat{\beta} = (-0.723, -0.699, -0.717)$ against -0.700 ; at $g = 0.20$, $0/20 \rightarrow 17/20$ ($p = 1.5 \times 10^{-5}$). At $g = 0.5$ the frozen policy still scores $15/20$ and the cell is a ceiling. Rotation gain faults are unidentifiable for this policy: no recorded step clears the excitation gate on r_x or r_z at any threshold, because the policy does not rotate the wrist. That is a property of the policy’s behaviour, not of the estimator.

6.4 BASELINES

Frozen and a **static oracle** correction equal to the true fault bound the problem. The informative baseline is the one that asks whether the plant model and M earn their place: **integral action on the raw motion error**, $\hat{f} \leftarrow \hat{f} + k_i(y_t - a_t)$, swept over five gains across two orders of magnitude. On the 0.15 translation fault its best setting ($k_i = 0.005$) reaches $7/20$ (35%) from a frozen $3/20$; higher gains fall to $0/20$ – $1/20$; ours reaches $19/20$ (95%) on the same cell. The baseline implicitly assumes achieved motion equals commanded action; the measured plant DC gain is 0.23, so with no fault at all its error is $-0.77a$ —at a command of 0.5 a phantom of 0.384 against a real fault of 0.150, $2.5\times$ the target and opposite in sign. Predicting $y \approx 0.23a$ instead of $y \approx a$ is the whole difference—and it is the plant model, not the update rule, that earns its place: matched estimators sharing the residual, M and projection (DOB, RLS, Kalman, a calibrated integral) score 19, 17, 17, 18 and 15 of 20 on this cell against 20 for an oracle, and a 5,424-rollout ALOHA tuning study finds no estimator beating another (Appendix B).

6.5 FAULTS BELOW THE CONTROLLER

Every fault above enters at the action interface. J-PARC’s enter in the actuator, where the fault-to-motion map depends on configuration through the Jacobian. We inject that class in the MuJoCo model of the LIBERO Panda, below the operational-space controller: a constant elbow torque (5 N m leaks through the controller as an 8–9 cm drift over 40 steps), added elbow friction, and an elbow lock (both invisible while holding, removing 3–10 cm of commanded motion). An estimate-only probe identifies each on the translation channels and nowhere else, the lock at 15–23 healthy-phantom standard deviations. Paired cells, $\pi_{0.5}$, `libero_spatial`, translation corrected, phantom subtracted:

joint fault (elbow)	n	frozen	corrected	fixed	broken	McNemar
torque bias 5 N m	40	20/40	32/40	15	3	0.0075
friction +10	20	11/20	14/20	6	3	0.51
friction +20	20	0/20	8/20	8	0	0.0078
lock ± 0.05 rad	20	0/20	0/20	0	0	—

The torque bias is repaired at a cost the Cartesian cells never paid, three regressions in forty: its estimate’s across-episode spread is 0.075, four times any Cartesian fault’s, because the offset the residual reports is right for the configuration it was measured in and wrong for the one the arm moves into. Heavy friction, which zeros the policy, is repaired to 40% with no regression; milder friction is not a claim ($p = 0.51$): friction acts only while the joint moves, so an estimate right during a reach pushes the wrong way when the arm pauses to grasp. The lock is identified (the estimate converges within 30 steps, reading the friction cell’s loss-of-motion signature), corrected, and not repaired. The difference is physical: friction is a force the controller overcomes by commanding more; a lock removes a degree of freedom no command restores. Identifiability is necessary for repair and not sufficient; the fault must be an input the plant still responds to. J-PARC’s offline residual reports gains under a lock, consistently: a residual trained on faulted rollouts can learn a different route through the free joints, and this law cannot. Across all seven Panda joints repair is joint-dependent: translation-only correction harms joint 5 ($12/20 \rightarrow 3/20$); a weighted full-channel allocator repairs it ($15/60 \rightarrow 41/60$) and breaks Object joint 3 ($16/20 \rightarrow 2/20$), an experimental option (Appendix B).

6.6 TWO FURTHER BACKBONES, UNCHANGED CALIBRATION

OpenVLA-OFT (7B, autoregressive backbone with an L1 action head) and NVIDIA GR00T N1.7 (3B, Cosmos VLM with a flow-matching DiT head, the official LIBERO finetune) are each served behind the same interface; every script, the plant and M are the $\pi_{0.5}$ ones.

cell (libero_spatial)	frozen	corrected	fixed	broken	McNemar
healthy control	20/20	19/20	0	1	1.0
rotation 0.10	0/20	11/20	11	0	9.8×10^{-4}
translation 0.15	0/20	14/20	14	0	1.2×10^{-4}
gain 0.20	0/20	17/20	17	0	1.5×10^{-5}

Nothing in the method was $\pi_{0.5}$ -specific. OFT is more fragile (frozen 0/20 on all three faults), recovers less on rotation (55% vs 85%; its r_y fault is only 23% identified where $\pi_{0.5}$ reached 87–100%), and the healthy control lost one episode to a sustained phantom four to ten times the usual size, which a confidence gate does not remove without delaying genuine repair by one to two seconds (§7). The three cells replicate on `libero_object` with the same calibration (0/20 $\rightarrow$ 10/20, 0/20 $\rightarrow$ 15/20, 0/20 $\rightarrow$ 19/20).

GR00T N1.7 is a third developer and a third architecture. Its healthy control is a null (18/20 both arms, 1 fixed, 1 broken; rotation phantom below 0.005). With no retuning: rotation 0.10, 0/20 $\rightarrow$ **14/20** (14 fixed, 0 broken, $p = 1.2 \times 10^{-4}$); translation 0.15, 2/20 $\rightarrow$ **15/20** (13/0, $p = 2.4 \times 10^{-4}$); gain 0.20 under the multiplicative law, 0/20 $\rightarrow$ **19/20** (19/0, $p = 3.8 \times 10^{-6}$, $\hat{\beta}$ within 2% of the true -0.80). Three families, 46 repaired, none broken. The under-identified r_y lags on all three backbones: a property of the plant and M , not of the policy. Across the three backbones on this cell, 42 of 60 paired episodes are repaired and none is broken.

6.7 A SECOND MANIPULATOR, IN JOINT SPACE

Bimanual ALOHA in `gym_aloha` driven by π_0 : fourteen absolute joint targets at 50 Hz, transfer-cube, 300-step episodes, healthy success 25%. The plant is identified on *position* from eight healthy rollouts ($R^2 = 0.989$ –1.000 on all joints), $M \approx I$. A +0.05 rad offset on the six left-arm joints displaces the gripper by 5.3 cm; the frozen policy scores 0/20 at every severity tested.

Identification transfers. Separation against the healthy control: 85–91% of the fault at 0.05 rad, 94–106% at 0.02 rad, leakage ≤ 0.007 rad onto uncorrected joints. **Repair does not, at first.** With continuous adaptation the task returns 0/20 at every severity, warm-started or cold, with or without an early-history defect we found and fixed. The reason is quantitative:

correction (0.02 rad fault, $n = 20$)	within-episode variation	mean residual	success
none (frozen)	—	2.12 cm	0/20
adaptive, updating throughout	0.43 cm	0.17 cm	0/20
0.019 rad (95%) applied frozen	0	0.11 cm	5/20
adapt episode 0, then hold , $n = 40$	0	0.12 cm	15/40 ($p = 6.1 \times 10^{-5}$, 0 broken)
healthy, no fault, $n = 40$ (law running: 14/40, 5 fixed, 8 broken, $p = 0.58$)	—	—	17/40

A static-correction sweep puts the task’s margin under 0.5 cm: a 90% correction held for the whole episode restores nothing, 100% restores the healthy rate. The updating law reaches a mean residual inside that margin and still fails, because its correction moves by 0.43 cm within an episode, the whole margin. The same number, applied stationary, repairs. *Identify, then hold* (adapt during one sacrificial episode, hold thereafter) recovers 15/40 from a frozen zero (held estimate 95–101% of the fault), the difference from the healthy arm’s 17/40 on the same seeds unresolved ($p = 0.75$); on a healthy arm the law holds a phantom below 0.001 rad and scores 14/40 against 17/40 (5 fixed, 8 broken, $p = 0.58$): no resolved effect, which is not the same as no effect. This is Proposition 2, invisible on LIBERO because a Cartesian reach tolerates centimetres of jitter.

6.8 A HUMANOID

Fourier GR1 upper body (two 7-joint arms, two 6-joint hands, a waist; 29 absolute joint targets at 20 Hz) in RoboCasa, driven zero-shot by GR00T N1.5, for which GR1 is a pretrained embodiment.

Plant from six healthy episodes ($R^2 \geq 0.998$ on the arm), M by direct step response (0.99 I on the arm block; the replay probe of §6.7 is contaminated on an arm in contact), constants from the measured residual. A +0.10 rad offset on the working arm zeros the policy on both tasks. The simulator’s scene generator is reseeded per episode so both arms see identical scenes. The estimate is the median over three episodes of the reach-window (steps 50–200) estimate: an episode’s end reads a contact standoff, 0.2 rad wrong on a wrist on two of three draws.

task, $n = 30$ paired	healthy	frozen	corrected	fixed	broken	McNemar	vs healthy
plate-to-plate	22/30	1/30	19/30	18	0	7.6×10^{-6}	$p = 0.55$
tray-to-plate	17/30	1/30	13/30	12	0	4.9×10^{-4}	$p = 0.45$

Held vectors are 93–104% and 73–126% of the fault per joint; on a healthy arm the law scores 19/30 vs 18/30 and 16/30 vs 15/30 (7 fixed, 6 broken each, $p = 1$); continuous adaptation reaches 8/30. Appendix G maps three magnitudes.

6.9 A SECOND SIMULATOR: WIDOWX IN SIMPLERENV

SimplerEnv (Li et al., 2024) runs in SAPIEN; GR00T N1.7’s Bridge finetune drives a WidowX on *put the spoon on the towel* with Cartesian pose increments on an accumulating target. Plant from ten healthy episodes, M the plant’s DC gain (the held-pose probe is wrong on z here), constants from the residual. A +0.005 offset on x, y, z , $2.5\times$ the residual, zeros the policy. The estimator is exact (median 0.0050, 0.0051, 0.0049), but the drift during its 27-step transient stays in the target: online adaptation reaches 6/20 against a healthy 13/20 (6 fixed, 0 broken), and 12/20 (12 fixed, 0 broken) with the gain raised to halve the transient. Identify over three episodes and hold: on the next 20 paired episodes healthy 14/20, frozen 0/20, corrected **14/20** (14 fixed, 0 broken, $p = 1.2 \times 10^{-4}$; vs healthy $p = 1.0$), the ALOHA and GR1 scheme.

7 WHAT DOES NOT WORK, AND WHAT WAS WRONG

Invisible faults. A proprioceptive bias damages nothing (15/15 frozen): this policy ignores its state input (§3), so the method’s blind spot coincides with the policy’s. A wrist-image shift damages the policy (6/12) and never reaches the residual (0.004 against 0.049). **A policy that cannot do the task; a plant that cannot be modelled.** `libero_90` fails 14 of 20 sampled tasks unfaulted, so there is nothing to restore; SimplerEnv’s Google robot under base OpenVLA (planner-interpolated 3 Hz deltas, $R^2 = 0.57$, healthy 2/10, a +0.02 offset at 20–60% of its residual signature) fails both conditions and no cell was run. **Harm on a healthy robot.** One OFT episode in twenty was lost to a sustained phantom acted on; a dwell gate removes most spurious openings, not that one. Three explanations we proposed were refuted by our own measurements and are kept with what killed them (Appendix C).

8 LIMITATIONS

Faults below the controller are repaired only when they remain inputs the plant responds to: a torque bias and heavy friction are, a joint lock is not (§6.5). A hardware stiffness loss is invisible in position and recoverable in effort (Appendix F): the sensor changes, not the law. Where the controller keeps the transient the deployable scheme is identify-and-hold, a weaker claim than online adaptation. Policy stochasticity gives ± 10 points at $n = 20$. We give propositions, not a recovery theorem. And every constant is a ratio of a measured scale that must be re-measured on a new plant: seven of our own failures came from not doing so.

9 CONCLUSION

A frozen VLA can be repaired within one episode by six numbers at its action interface, given a plant model and a sensitivity matrix identified once on healthy data. One calibration serves four suites and three backbones, the two conditions that govern success are measurable in advance, and the failures are named. The result still owed is the one on a real robot.

AI USE STATEMENT

Generative AI (Claude, Anthropic) was used throughout this work as a coding and writing assistant under the authors’ direction: it wrote and edited the experiment scripts, the analysis code, the running project record from which this paper is built, and this manuscript’s text; the authors set the research questions, chose the experiments, ran or supervised every run on their own hardware, and reviewed the outputs. All numbers in the paper are recomputed from stored per-episode outcomes by a stdlib-only script, so no reported figure rests on model-generated text. Three mechanistic explanations proposed during the work, some AI-generated, were refuted by our own measurements and are reported as such (§7). Related work was verified against the cited preprints. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

Every number above is recomputed from stored per-episode outcomes by `openpi/mcnemar.py` with nothing installed; the anonymised repository holds those outcomes, the plant and sensitivity calibrations, every script named in the paper, and `aggregate_tally.py`, which computes the abstract’s paired-episode tally from the stored outcomes and writes a manifest naming every included and excluded file. Experiments need `openpi` at commit 15a9616, the `pi05_libero` checkpoint, and one GPU. The two-virtualenv split, the OpenVLA-OFT and ALOHA environments, and the three pins that were not obvious are in `SETUP.md`. The law’s constants are stated as ratios of measured scales (§4), which is what made the calibration transfer across suites and backbones without retuning.

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A THE CHANNEL GATE FROM HEALTHY DATA ALONE

A gate from healthy data alone. The separation test above needs a faulted rollout. A preregistered alternative uses none: twenty healthy estimate-only episodes give, per channel, the mean b_i and across-episode standard deviation sd_i of the estimate averaged over the last 50 steps; channel i is corrected at a step iff $|\hat{f}_i - b_i| > 3 sd_i$, re-evaluated every step. The translation phantom is wide (3 sd of 0.12 on x and z , above the 0.05 fault), which is the healthy-data reason translation is untrustworthy on this plant. On the headline scenarios: frozen 9/20, gated 19/20, 10 fixed, 0 broken, $p = 0.002$ (rotation-only mask: 8/20 $\rightarrow$ 18/20). Occupancy after step 20: r_x, r_z open on every step, r_y on 28% (its estimate settles at 0.021 against a threshold of 0.038), x 17%, y 36%, z 20%; the registered prediction had translation under 20%, which y refutes without cost. Healthy control with the gate: 20/20 $\rightarrow$ 19/20 (0 fixed, 1 broken, $p = 1$); the gate opens on 30% of healthy steps because a threshold calibrated on a 50-step mean is read on the instantaneous estimate, the registered $< 5\%$ was wrong, and one healthy episode timed out. Record §37, `prereg_records/PREREG_HEALTHY_GATE.md`.

B TWO STUDIES FROM THE DUAL-TRACK AUDIT

The repository’s second author audited the paper against its stored data (two independent tracks, 25 of 26 headline numbers reproduced, six summary claims corrected in this version, the McNemar instrument’s absolute tolerance replaced by a relative one) and ran two studies whose results this paper reports as they stand.

All-joint physical faults. 1,440 Panda rollouts: seven +5 N m joint torques plus healthy, three LIBERO suites, 20 scenarios per cell, arms off / legacy translation-only / weighted full-channel allocation; and 780 ALOHA rollouts over all twelve arm joints. The weighted allocator changes both the permitted channels and the regularisation; it does not isolate either.

condition	off	legacy	weighted	comparison
Panda joint 5, three suites	36/60	15/60	41/60	primary $p = 2.56 \times 10^{-6}$
Panda Object, joint 3	8/20	16/20	2/20	Bonferroni-24 $p = 0.012$
Panda Goal, healthy	19/20	3/20	19/20	Bonferroni-24 $p = 0.0007$
ALOHA, all faulted cells	60/240	70/240	66/240	descriptive total
ALOHA, healthy	11/20	8/20	6/20	unresolved after Bonferroni-13

Legacy translation-only correction harms healthy Goal (19/20 $\rightarrow$ 3/20) and joint 5; the weighted allocator repairs joint 5 and breaks Object joint 3. Repair below the controller is joint- and task-dependent, and the allocator is experimental.

Matched estimators on ALOHA. Six estimator families sharing the plant model, M and projection, tuned with a fixed budget on separate seeds (fit 3100–3129, search 3200–3203, validation 3300–3307) and confirmed on 30 untouched seeds (3400–3429), 5,424 rollouts in all. Balanced score = half healthy success plus half the mean of seven fault rates.

method	healthy	faulted	balanced
this law	16/30	98/210	50.0%
RLS	17/30	63/210	43.3%
DOB	14/30	79/210	42.1%
Kalman	12/30	71/210	36.9%
composite (prediction + tracking)	12/30	67/210	36.0%
calibrated integral	11/30	57/210	31.9%
off	15/30	48/210	36.4%
oracle (unranked)	13/30	103/210	46.2%

The ordering is numerical, not established: composite minus Kalman is -0.95 points with a whole-seed bootstrap interval $[-9.5, +7.6]$, and no contrast survives correction. Everything with the plant model repairs; nothing in the update rule is decisive.

C REFUTED EXPLANATIONS, KEPT

That identifiability follows from healthy action statistics (translation identifies a single-axis fault at 79%); that the plant is nonlinear on translation (M is magnitude independent to 14%); that translation correction contributes nothing (20% $\rightarrow$ 95% at 0.15).

D HELD-OUT CALIBRATION

Preregistered (prereg_records/PREREG_HELDOUT_CALIBRATION.md): plant and M re-identified from ten healthy episodes at initial states from 25 upward, disjoint from the evaluated 45–48, and the four headline cells rerun with nothing else changed.

The held-out M agrees with the shipped one on rotation (within 8%) and is 19–21% lower on y and z , refuting the registered 15% band on translation: the sensitivity depends on the configuration the probe runs from. Condition number 3.1 against 3.0. The held-out plant fits as well (R^2 on x, y, z 0.95/0.98/0.96 against 0.96/0.98/0.98) and better on r_x (0.81 against 0.52, ten episodes against three); r_y is the worst-fit channel in both logs (0.33–0.42), which is the same channel whose estimate lags on all three backbones and which the healthy gate of Appendix A keeps mostly closed.

Cell results are the right-hand column of Table 1: 18, 15, 28 and 7 corrected against 18, 16, 29 and 15. The registered bound (each cell within three) holds on three suites and fails on libero_10 by eight. Repair there is still real and significant (0/40 $\rightarrow$ 7/40, 7 fixed, 0 broken, $p = 0.016$) and half what the overlapping calibration delivered. Pooled 34/120 $\rightarrow$ 68/120, 37 fixed, 3 broken. We report both columns rather than the better one: a deployment that calibrates on the robot it will run on gets the left column, and one that calibrates elsewhere should expect the right.

E ABLATION OF THE LAW’S CONSTANTS AND THE CALIBRATION SIZE

Headline cell ($\pi_{0.5}$, libero_spatial, uniform +0.05, rotation-only correction), $n = 20$ paired per row, one constant changed per row from the reference ($\gamma = 0.08$, $\rho = 0.15$, deadzone 0.008, plant fitted on three healthy episodes, 285 steps). Last column: rotation estimate at episode end (last-50-step mean) against a true 0.05 on each axis.

setting	frozen	corrected	fixed	broken	McNemar	$\hat{f}_{rx}$, $\hat{f}_{ry}$, $\hat{f}_{rz}$
reference	8/20	18/20	10	0	0.0020	0.044, 0.020, 0.044
$\gamma = 0.02$ ($\frac{1}{4}\times$)	8/20	17/20	9	0	0.0039	0.036, 0.016, 0.036
$\gamma = 0.32$ ($4\times$)	7/20	18/20	11	0	0.00098	0.044, 0.019, 0.044
$\rho = 0.05$ ($\frac{1}{3}\times$)	11/20	14/20	5	2	0.45	0.027, 0.011, 0.027
$\rho = 0.50$ ($3\frac{1}{3}\times$)	8/20	20/20	12	0	0.00049	0.048, 0.022, 0.049
deadzone 0	8/20	19/20	11	0	0.00098	0.044, 0.018, 0.044
deadzone 0.03 ($\approx$ residual scale)	8/20	17/20	9	0	0.0039	0.031, 0.016, 0.032
plant from 1 healthy episode (75 steps)	10/20	17/20	8	1	0.039	0.027, 0.008, 0.044
plant from 2 healthy episodes	8/20	17/20	10	1	0.012	0.045, 0.021, 0.043

The gain is flat across a $16\times$ range: it sets the length of the transient, which at a quarter of the reference still ends well inside a 220-step episode, and the final estimate is unchanged. The deadzone is flat until it reaches the residual’s own scale, where it eats signal. The normaliser is the one constant that decides the outcome, in the direction the law’s fixed point $f/(1+|r|^2/\rho^2)$ predicts: $\rho = 0.05$ against a measured residual of 0.034 puts the estimate at 55% of the fault and the cell becomes a null with two regressions; $\rho = 0.50$ removes the bias and the cell reaches 20/20. The only setting that fails is the one chosen below the measured residual scale, which is the content of the “constants as ratios of measured scales” rule in §4. One healthy episode, 75 control steps, is enough for the plant: 17/20 with one rotation axis under-identified and the other two carrying the repair; two episodes recover the full estimate.

F HARDWARE PROTOCOL ON THE X2 HUMANOID

On logs from an X2 humanoid running a learned box-pickup policy, a real actuator-authority fault (stiffness, damping and feedforward scaled together, 0.1–1.3 across 134 rollouts) is invisible in joint position and recoverable in effort ($r = +0.84$); the plant is run-varying on load-bearing joints, so calibration must be within-run and in motion; the detectability floor is 0.5–1.3° of joint offset. The paired repair experiment on the robot is pending robot time and is the one result this paper does not yet contain.

G HUMANOID RECOVERABILITY MAP

Plate-to-plate, right-arm offset at three magnitudes, paired on identical scenes, identify over three episodes then hold (§6.8), $n = 30$ per arm:

offset	healthy	frozen	corrected	fixed	broken	McNemar	held estimate
0.05 rad	19/30	7/30	24/30	17	0	1.5×10^{-5}	90–108%
0.10 rad	22/30	1/30	19/30	18	0	7.6×10^{-6}	93–104%
0.20 rad	25/30	0/30	21/30	21	0	9.5×10^{-7}	94–99%

Every cell is at its healthy rate ($p = 0.18, 0.55, 0.29$ against the healthy arm) with no regression in 90 paired episodes; at 0.20 rad (11.5° on every joint of the working arm) the frozen policy never succeeds. The identification stays within 10% across a fourfold range of magnitude, with no sign of the superposition boundary the Panda shows at its largest uniform fault (§6.3).

Time-varying faults on the humanoid. Same cell, $n = 20$ paired, amplitude 0.10 rad. A ramp to full over 60 steps is constant thereafter, and the held estimate serves it: 1/20 $\rightarrow$ 11/20 (10 fixed, 0 broken, $p = 0.002$; healthy 13/20, $p = 0.73$). Continuous adaptation repairs the ramp significantly

but below healthy (9/20, $p = 0.021$ against frozen, $p = 0.039$ against healthy 16/20), lagging the rise by about 30 steps and then continuing to move. The two profiles that never settle are the boundary: a slow non-zero-mean oscillation reaches 6/20 and an intermittent fault 4/20, neither significant against frozen at $n = 20$. On the Panda the same profiles were repaired (§6.3) because a Cartesian reach tolerates a correction that is briefly wrong; a dexterous-hand grasp does not, which is Proposition 2 on the axis it names: a fault that keeps moving cannot be given a stationary correction.

H RECOVERABILITY-MAP CELLS AT $n = 40$

The two cells of §6.3 that were significant only at the exact test's floor or just above it were rerun at $n = 40$ (same configuration, initial states 45–48). Rotation 0.05: 27/40 $\rightarrow$ 39/40, 12 fixed, 0 broken, $p = 4.9 \times 10^{-4}$. Translation 0.10: 30/40 $\rightarrow$ 38/40, 9 fixed, 1 broken, $p = 0.021$. Every map cell that showed an effect at $n = 20$ is individually significant; uniform 0.15 is the superposition boundary and is not.